SPACE FOR INNOVATION

A powerhouse in space technology, private players are answering ISRO's call. More innovation is needed in other fields

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LONG BEFORE ECONOMIC prosperity, India had invested in space technology and atomic research—steps that have now made it a giant in these fields. Now, realising it as a key element of economic growth, great strides are being made in research in other areas of science too. India is one of the top five nations in space exploration. Gaganyaan, the Indian human spaceflight programme, an ambitious and difficult mission, is on track. The human space programme is a major milestone for India. The challenge here, as with India's space programme through the decades, is technology, for India has had to develop them largely on its own.

The Indian Space Research Organisation (ISRO) alone cannot create a larger space ecosystem. So, around 50 entrepreneurial ventures have stepped up to build and launch rockets, satellite constellations and develop applications. Public-private partnership is evolving rapidly in the space sector.

Under the ministry, use of space technology for efficient governance is growing as well. Property records are being prepared by surveying over 250,000 panchayats under the Svamitva scheme using a combination of survey grade drones and a Continuously Operated Reference Stations (CORS) network. The maps thus generated are highly accurate. When completed, it will be useful in implementing government schemes.

India now ranks 46 among 50 countries in the Global Innovation Index (GII)—an improvement from the 48th position in 2020. The ministry, led by Jitendra Singh, is promoting research parks that house technology business incubators, which would promote innovation till they become commercial ventures.

Singh is also in charge of the ministry of personnel. The ministry has facilitated the lateral entry of qualified individuals in the bureaucracy, apart from those from the states aspiring to work for the Union government and who are in the all India services, including the IAS and IPS. This enables the filling of vacancies of mid-level managers in ministries and results in faster decision-making. Online recruitment for some administrative posts has also been introduced. ■

WHAT'S DONE

- New Space India Ltd, ISRO's commercial arm, set up for industry involvement
- Space sector opened to private players, start-ups
- In FY21, the science and technology sector added 1,497,501 employees, becoming India's top employment generator

CHALLENGES AHEAD

- Investment required in nuclear energy sector as India tries to expand capacity
- More research needed in nanotechnology, which can transform pharma industry
- Laws and guidelines needed for IN-SPACe, a nodal agency of all space activity